Hybrid randomized clinical trials incorporating external controls: assumptions and related recommendations

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https://github.com/foucher-y/talks

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Plan

Augmenting randomized clinical trials (RCT) with external controls

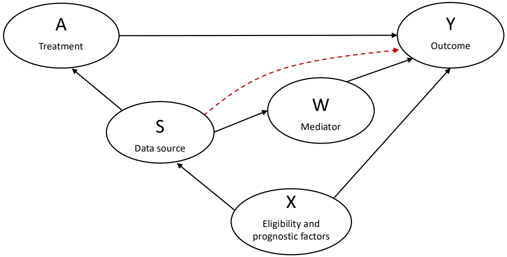
- RCTs represent the reference protocol for treatment evaluations.
- For large sample size, the comparability of the arms and the power acceptable.
- Number of trials are based on small sample sizes (median at 75 patients in ICU).[1]
- Novel methods aim to increase the power by using internal and external controls.
- It calls for trade-off between:
  - decreasing the random bias, and
  - increasing systematic bias.

How to reduce the risk of systematic bias?

[1] Anthon et al. (Journal of Clinical Epidemiology, 2019)

Notations and directed acyclic graph[2]

- $Y = (0, 1)$: the binary outcome.
- A: the treatment arm.
  - $A = 1$ for the experimental group.
  - $A = 0$ for the control group.
- S: the data source.
  - $S = 1$ for the internal RCT.
  - $S = 0$ for the external population.
- X: the causes of both Y and S.
- W: the mediators of being included in the RCT.



[2]Dang et al. (arXiv, 2023)

Estimand and targeted population

- Our objective is to estimate the average treatment effect (ATE) in the target population of the RCT.
- What would have been obtained from the RCT with a large sample size without augmentation with external controls?

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)|S = 1]$$

  - $Y(0)$ is the counterfactual outcomes under the control treatment.
  - $Y(1)$ is the counterfactual outcomes under the experimental treatment.
- When **three assumptions for causal identifiability** are met, one can estimate the ATE from observations:[3]

$$\text{ATE} = \mathbb{E}[Y|A = 1] - \mathbb{E}_{X|S=1}[E(Y|A = 0, X)]$$

[3] Valancius (Biometrics, 2024)

Plan

Assumption #1

- Positivity violation occurs when it is certain that patients are allocated to a group.
- Because of the randomization, this condition holds for individuals in the RCT.
- Therefore, positivity in treatment allocation is respected if the external population have a non-zero probability to be included in the RCT.

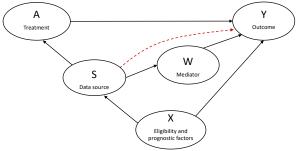
Positivity in RCT-recruitment (A1):

The external population have a non-zero probability to be included in the RCT.

$$\Pr(S = 1 | X = x, A = 0) > 0 \text{ for all } x \text{ with } p(x | A = 0) > 0$$

Assumption #2

- The factors X may represent a source of bias.
- For instance, elderly patients are often under-represented in a RCT. It may lead to an overestimated effect of the experimental treatment.
- It calls for the absence of measuring X for correcting the bias by modelling.



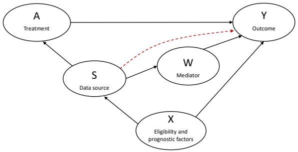
No unmeasured confounders (A2)

The eligibility criteria and prognostic factors X are measured.

Assumption #3

- The average outcome for controls given factors X is the same in the RCT and external population.

$$\mathbb{E}(Y|X, A = 0, S = 1) = \mathbb{E}(Y|X, A = 0, S = 0)$$



- It can be violated for different schedules of visits in a trial.
- The follow-up is an example of a mediator W.
- The measurement of W can further help in correcting the related bias.

No direct effect on the outcome of being in the trial (A3)

There is no residual effect of S on Y conditional on the confounders X and mediators W (who must be measured).

Recommendations to prevent violation of the assumptions by design

Assumption 1 Positivity in RCT-recruitment	**1.1.** Inclusion and exclusion criteria of the external and internal subjects are similar. **1.2.** External controls come from a concomitant or recent study. **1.3.** The external study consists for a large part of the same or similar centers.
Assumption 2 No unmeasured eligibility and prognostic factors X	**2.1.** Factors X are available in both the internal and external data. **2.2.** Factors X are measured similarly across the internal and external studies. **2.3.** Factors X should include social and lifestyle dimensions.
Assumption 3 No direct effect on the outcome	**3.1.** Baseline of the controls is identical in the RCT and the external study. **3.2.** External controls receive the same treatment as in the control ones. **3.3.** Patients in the RCT have a follow-up as close as possible as the routine care. **3.4.** Outcome Y are measured similarly in the RCT and the external study. **3.5.** Mediators W are measured similarly in the RCT and the external study. **3.6.** Missing data on the variables (X, W, Y) is rare and (completely) at random.

Plan

Context

Causal assumptions and prevention of their violations

Data-driven detection of violations

Conclusion/References

Treatment regression for evaluating the positivity in recruitment (A1)

- Description to check the overlap of the characteristics X between the internal and external controls.
- Estimation of a logistic regression (or another predictive model) with S as the outcome conditional on X to predict the source propensity scores by using all the internal and external patients.

$$\hat{p}_i = \hat{\Pr}(S = 1 | X_i = x_i) \text{ for a patient } i$$

- Identify the external patients with $\hat{p}_i < \epsilon$ (for instance $\epsilon = 0.05$)

In case of positivity violation, the investigators can update the inclusion criteria of the external population to recover positivity in RCT-recruitment.

Outcome regression for evaluating the absence of direct effect of being in the RCT (A3)

- The respect of A3 results in similar predicted outcomes for controls regardless the source:

$$\Pr(Y = 1|X, S = 0, A = 0) = \Pr(Y = 1|X, S = 1, A = 0)$$

- One can model $\Pr(Y = 1|X, S)$ among the controls for testing residual effect of S
- One can model $\Pr(Y = 1|X)$ among the external controls and investigate its calibration among the internal controls.

In case of violation, one can further explore if the direct effect can be explained by W: $\Pr(Y = 1|X, W, S = 0, A = 0) = \Pr(Y = 1|X, W, S = 1, A = 0)$

Required sample size for such a calibration

- In our context with a binary outcome, one can propose checking that $O/E = 1$:
  - $O = \sum_i Y_i$ is the number of observed events among the internal controls, and
  - $E = \sum_i \hat{\Pr}(Y_i = 1 | X = x_i, S_i = 1, A_i = 0)$ is the expected number thanks to the outcome regression.
- The required sample size for such a calibration can be estimated by:[4]

$$N_{0c} = (1 - \pi_0)/(\pi_0 SE_{\alpha/2,I}(\log(O/E)^2)$$

  - π_0 is the expected proportion of events ($Y = 1$) among controls, and
  - $SE_{\alpha/2,I}(\log(O/E))$ is the standard deviation associated with the width I of the targeted (1-α)% confidence interval (CI) of $\log(O/E)$

[4] Riley et al. (Statistics in Medicine, 2021)

Required sample size for a superiority trial

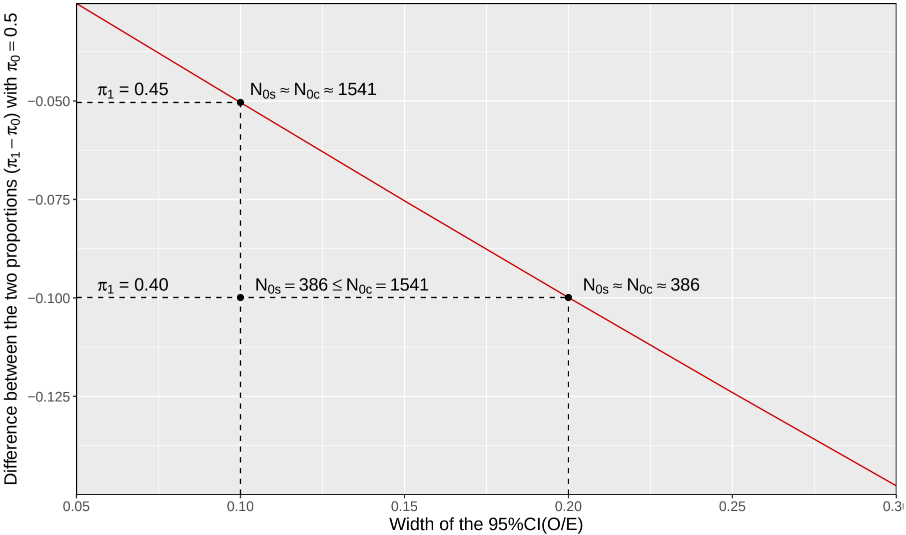
- For a usual 1:1 superiority RCT, the required sample sizes in each arm can by estimated by:

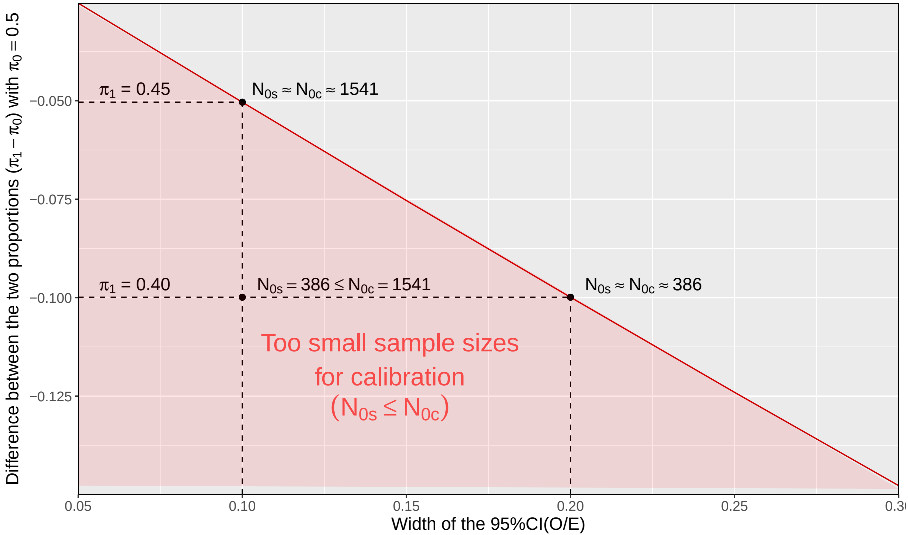
$$N_{0s} = N_{1s} = \frac{(z_{\alpha/2} + z_{\beta})^2 \{\pi_0(1-\pi_0) + \pi_1(1-\pi_1)\}}{(\pi_1 - \pi_0)^2}$$

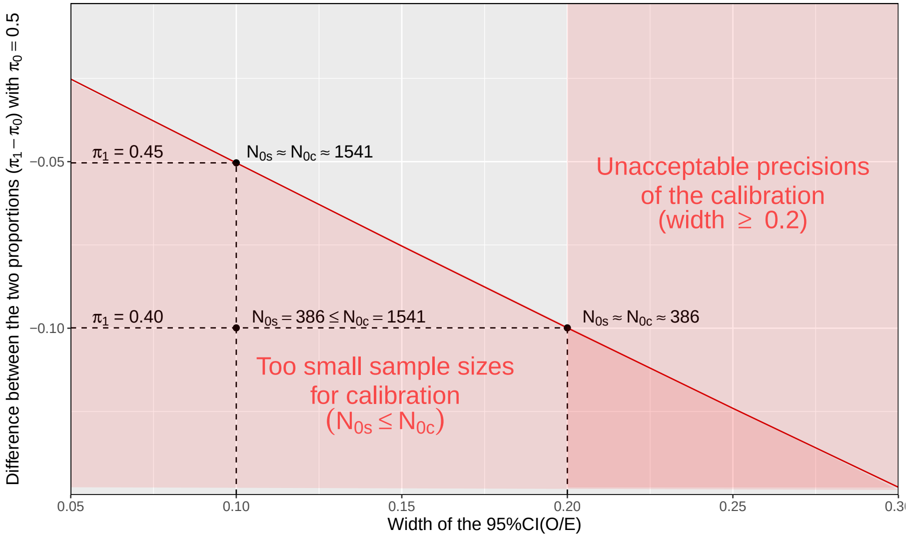
  - π_1 is the expected proportion of events $(Y = 1)$ among the experimental arm,
  - α is the type-I error, and
  - β is the type-II error.

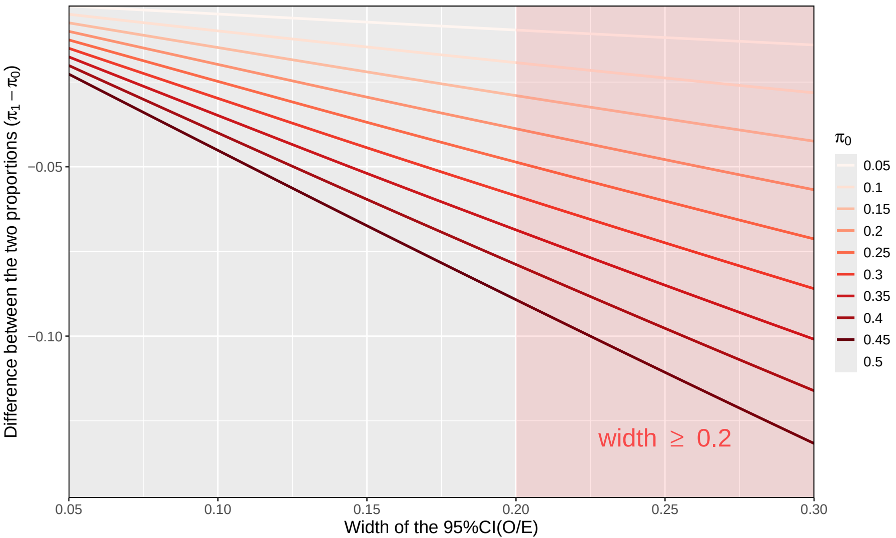
Two illustrative examples with $(\alpha, \beta) = (0.05, 0.20)$

- Example 1:
  - $(\pi_0, \pi_1) = (0.50, 0.45)$
  - Width of the $95\% CI(\log(O/E))$ equals to 0.20
  - $SE_{0.025,0.20}(\log(O/E)) \approx 0.051$
  - Sample size of controls for the usual 1:1 superiority trial: $N_{0s} \approx 1562$ patients.
  - Sample size of controls for calibration: $N_{0c} \approx 386$ patients.
  - **The effective of internal controls will allow for investing the calibration.**
- Example 2:
  - Same data but an higher effect: $(\pi_0, \pi_1) = (0.50, 0.35)$
  - Sample size of controls for the usual 1:1 superiority trial: $N_{0s} \approx 170$ patients.
  - **The effective of internal controls will not allow for investing the calibration.**
- Example 3:
  - $(\pi_0, \pi_1) = (0.50, 0.40)$: $N_{0s} \approx N_{0c} \approx 386$ patients.


Difference between the two proportions ($\pi_1 - \pi_0$) with $\pi_0 = 0.5$
Width of the 95%CI(O/E)
$\pi_1 = 0.45$
$N_{0s} \approx N_{0c} \approx 1541$
$\pi_1 = 0.40$
$N_{0s} = 386 \leq N_{0c} = 1541$
$N_{0s} \approx N_{0c} \approx 386$
−0.050
−0.075
−0.100
−0.125
0.05
0.10
0.15
0.20
0.25
0.30


π_1 = 0.45
$N_{0s} \approx N_{0c} \approx 1541$
π_1 = 0.40
$N_{0s} = 386 \leq N_{0c} = 1541$
$N_{0s} \approx N_{0c} \approx 386$
Too small sample sizes
for calibration
($N_{0s} \leq N_{0c}$)
Difference between the two proportions ($\pi_1 - \pi_0$) with $\pi_0 = 0.5$
−0.050
−0.075
−0.100
−0.125
0.05
0.10
0.15
0.20
0.25
Width of the 95%CI(O/E)


Difference between the two proportions ($\pi_1 - \pi_0$) with $\pi_0 = 0.5$
−0.05
−0.10
$\pi_1 = 0.45$
$N_{0s} \approx N_{0c} \approx 1541$
$\pi_1 = 0.40$
$N_{0s} = 386 \leq N_{0c} = 1541$
$N_{0s} \approx N_{0c} \approx 386$
Unacceptable precisions of the calibration (width $\geq$ 0.2)
Too small sample sizes for calibration ($N_{0s} \leq N_{0c}$)
0.05
0.10
0.15
0.20
0.25
Width of the 95%CI(O/E)



Difference between the two proportions ($\pi_1 - \pi_0$)
−0.05
−0.10
0.05
0.10
0.15
0.20
0.25
0.30
Width of the 95%CI(O/E)
width ≥ 0.2
π_0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

Plan

Summary of the key points for such an hybrid trial

- External controls represent an opportunity, but 3 assumptions should be respected.
- We proposed a bundle of 12 recommendations which can guide the study design.
- It underlined the importance of prospectively designing both the internal and external sources as close as possible:
  - Retrospective registries or medico-administrative databases as external sources should be used with caution.
  - The prospective collection of the internal data must be designed according to the methods used for external controls.
- Paradox: While external controls may be used when the internal sample size is small, exploring the respect of A3 necessitates large sample size.
- A3 cannot be investigated for full-external controlled trials...

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